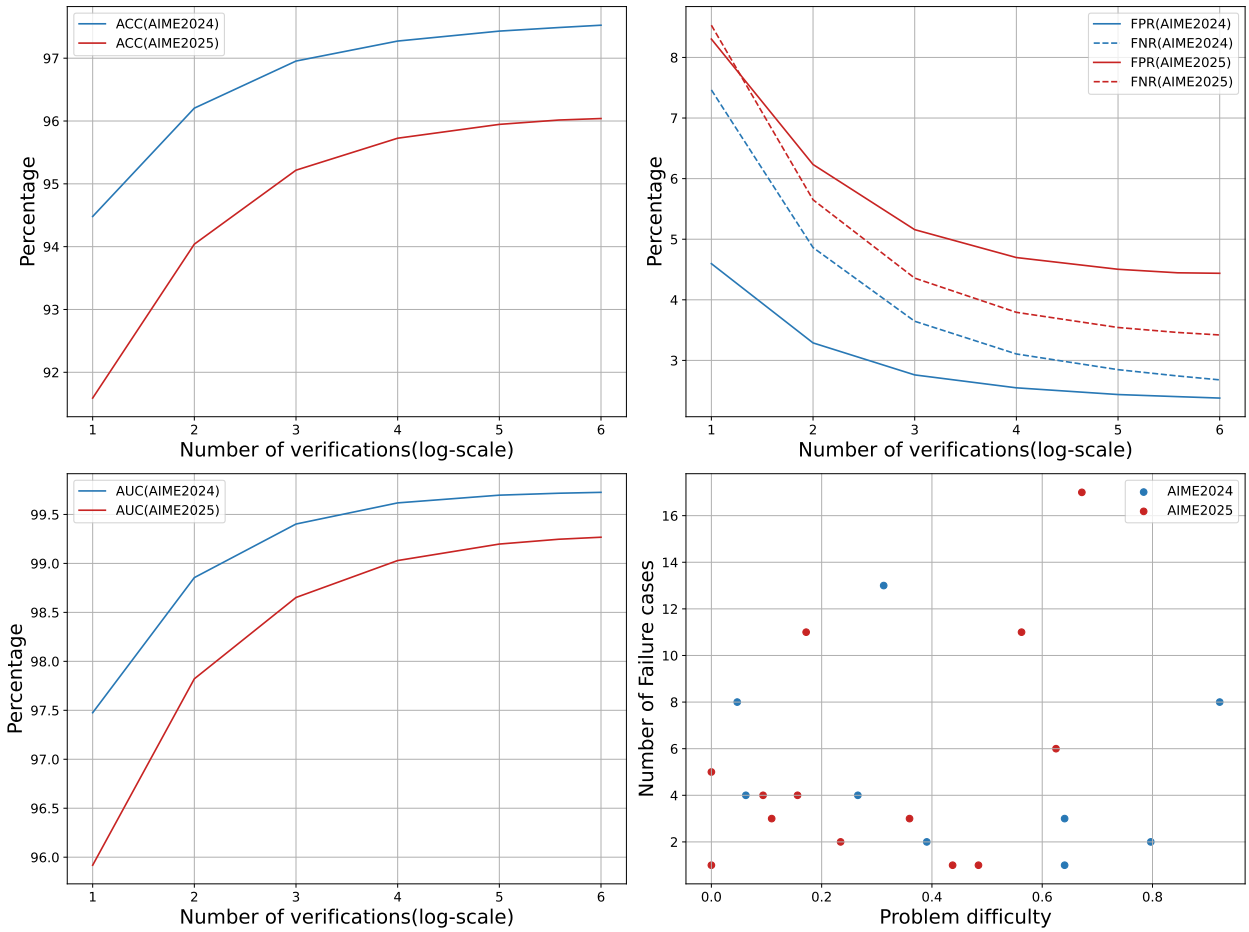

Figure 3 The inference-time scaling of verification ability on problem solutions in AIME2024 and AIME2025. **Top-left:** We show the accuracy of Heimdall when we sample multiple verification responses and make the judgment by majority voting. **Top-right:** We show the decreasing false-negative rate(FNR) and false-positive rate(FPR) as we scale up verification responses with majority voting. **Bottom-left:** We calculate the average score of verification responses and draw the AUC along each number of responses. **Bottom-right:** We collect the verification failure cases on every math problem and draw the relation between the difficulty of the problem and the number of verification failures, which reveals that the verification difficulty may not necessarily correlate with the difficulty of the original problem.

...

Problem
Let the sequence of rationals $x_1, x_2, \ldots$ be defined such that $x_1 = \frac{25}{11}$ and

$$x_{k+1} = \frac{1}{3}\left(x_k + \frac{1}{x_k} - 1\right).$$

x_{2025} can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n. Find the remainder when $m + n$ is divided by 1000.

Solution
...

Verification
<think>
Okay, so I need to figure out if the given solution to this problem is correct. Let me read through it again to make sure I understand.
The problem is: ...
The solution provided has four key steps. Let me go through each step and verify.
Step 1: They let $x_k = a_k/b_k$ with a_k and b_k coprime. Then they write the recurrence relations as $a_{k+1} = a_k^2 + b_k^2 - a_k b_k$ and $b_{k+1} = 3a_k b_k$.
Wait, let's verify that. Starting from $x_{k+1} = (1/3)(x_k + 1/x_k - 1)$. If $x_k = a_k/b_k$, then $1/x_k = b_k/a_k$. So $x_k + 1/x_k = (a_k^2 + b_k^2)/(a_k b_k)$. ... Okay, so step 1 seems correct. [Forward checking]
...
Wait let me confirm with S_3: $a_3 = a_2^2 + b_2^2 - a_2 b_2$. Wait $a_2 = 471$, $b_2 = 825$. So $a_3 = 471^2 + 825^2 - 471 * 825$.
... [Backward checking]
...
</think>
The solution correctly arrives at the conclusion despite some initial confusion in their explanation. They properly computed each part modulo 1000, leading to the correct final result.
Answer: 1

Table 2 A case of verification on hard problems. The verification process contains both forward checking, i.e., checking if each step is correct and backward checking, i.e., checking if a conclusion matches constraints, as is shown in the position of the red texts.

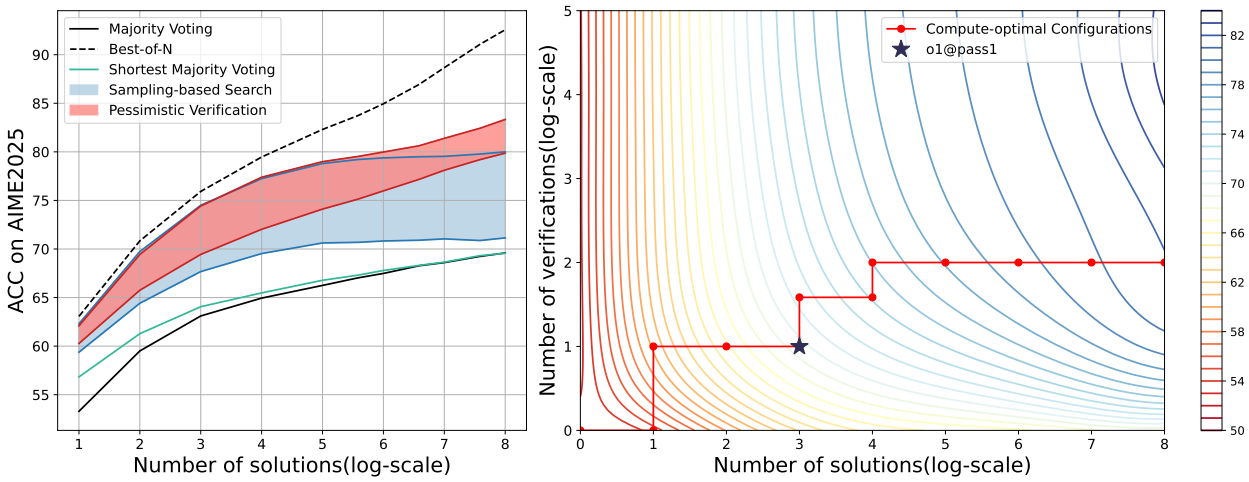

Figure 4 The inference-time scaling of problem solving with Heimdall. The two figures show the accuracy on AIME datasets as the number of solutions scales up. **Left:** the problem solving accuracy on AIME2025 dataset scales with the number of solutions. The colored shaded area represents the area covered by the accuracy curves of a selection algorithm as the number of verifications increases from 1 to 64. **Right:** the contour map of the accuracy of Pessimistic Verification as the number of solutions (x-axis) and the number of verifications (y-axis) increase. The red curve indicates the optimal configurations within various overall compute budgets.

time scaling in two dimensions, i.e., the number of solutions N and the number of verifications on each solution M. Considering the huge computational cost, we only evaluate on AIME2025, with $N \in [2, 256]$ and $M \in [1, 64]$. Similar to the analysis of verification accuracy, we first sample $N = 256$ solutions for each problem and $M = 64$ verifications for each solution as the complete data and then randomly select a subset to evaluate each scaling algorithm under a specific compute budget (M, N). We repeat the sampling for 2048 time to get a stable average score. We set $\alpha = 0.1$ in Pessimistic Verification and also evaluate the other three selection algorithms as follows.

Majority Voting Majority voting is one of the most commonly used inference time scaling methods. It first categorize the solutions, e.g., by the final answers for math problems. It simply select the category that contains the largest number of solutions in it. As fore-mentioned, majority voting can be seen as a special case of Pessimistic verification, where α is large enough to overshadow the signal of verification.

Shortest Majority Voting. The recent work [31] observes a length bias that the correct solutions are often shorter than incorrect ones for the same questions. Suppose the answer a_i occurs c_i times in the sampled responses and the average length of responses with the answer a_i is l_i, the voting score for a_i is

$$s_i = \frac{c_i}{l_i}$$

Sampling-based Search. The work [33] leverages a commercial LLM as the verifier, and scales the inference-time computation on the number of sampled solutions and the number of verifications. During the selection, it calculates the average verification score of each solution and selects the solution with the largest score. Note that it does not group the solutions based on their answers, which is different from the special case of Pessimistic verification where α equals zero.

Tie-breaking rules. The selection algorithm may encounter a tie situation, where multiple options have the same score. In principle, one can introduce another model to compare which option is better, but for simplicity, we leverage the length prior to break the tie, namely, selecting the option with the shortest average solution length.